EE1-06 Kin K. Leung SOLUTIONS

2016 Intro. to Signals & Communications Exam

1. a. i. $$F(\omega) = \int_{-\infty}^{\infty} f(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-b}^{b} a\, e^{-j\omega t}\, dt$$

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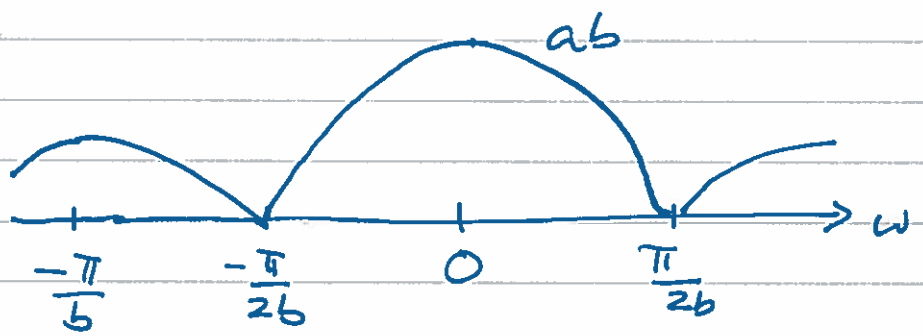
$$= a \cdot \frac{1}{-j\omega} \cdot \frac{e^{-j\omega b} - e^{j\omega b}}{2}$$

$$= a \cdot \frac{1}{-j\omega} \cdot \frac{-2j \sin(\omega b)}{2}$$

$$\Rightarrow F(\omega) = \frac{a \sin(\omega b)}{\omega}.$$

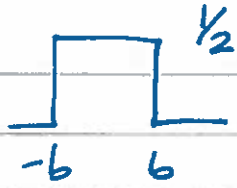
$$F(\omega) = ab \cdot \mathrm{sinc}(\omega b)$$ (4)

ii)



(3)

phase $\angle F(\omega) = 0$ because $F(\omega)$ is real

1. a. iii. $f(t) = \lim_{b \to 0}$ 

(2)

$\Rightarrow f(t) = \delta(t)$ ∵ the area is unit and the width tends to zero

In this special case

$$F(\omega) = \mathcal{F}[\delta(t)] = 1.$$

iv. $\mathcal{F}[f(t-t_0)]$

$$= \int_{-\infty}^{\infty} f(t-t_0)\, e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} f(t-t_0)\, e^{-j\omega(t-t_0)} \cdot e^{-j\omega t_0}\, dt$$

$$= e^{-j\omega t_0} \int_{-\infty}^{\infty} f(t-t_0)\, e^{-j\omega(t-t_0)}\, dt$$

$$= e^{-j\omega t_0} \cdot F(\omega)$$

$$\Rightarrow \hat{F}(\omega) = e^{-j\omega t_0} F(\omega)$$

(3)

1. b. i. $\omega_0 = \frac{2\pi}{T_0}$ (1)

ii. a_0 is the dc component of the signal $g(t)$. (1)

iii. a_n reflects the degree of how $g(t)$ is similar to $\cos(n\omega_0 t)$

b_n : similarity w/ $\sin(n\omega_0 t)$ (2)

iv. If $g(t)$ is odd function of t,

$$g(-t) = -g(t)$$

So,

$$g(-t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(-n\omega_0 t)$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) - \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

Since $g(-t) = -g(t)$

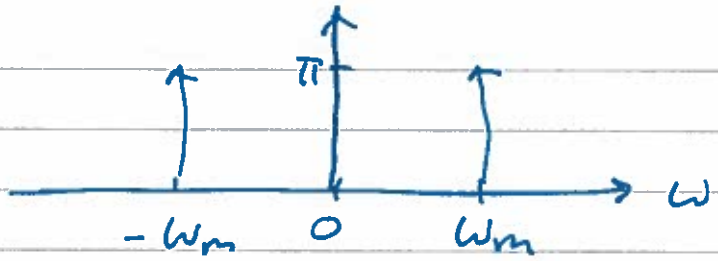
$$\Rightarrow a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) - \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t)$$

$$= -a_0 - \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) - \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t$$

$$\Rightarrow a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) = 0$$ (3)

$$\Rightarrow a_0 = 0, \; a_n = 0 \; \forall \text{ all } n = 1, 2, \ldots \infty$$ (3)

1. c. i.



(2)

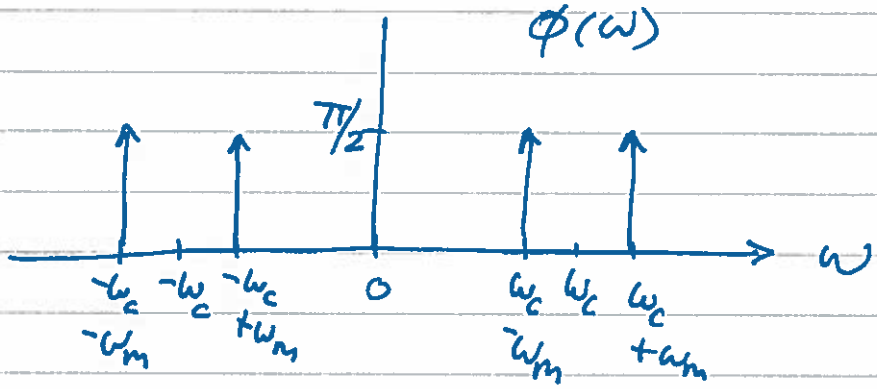
ii. $\phi(t) = m(t)\cos(\omega_c t)$

$= \cos(\omega_m t).\cos(\omega_c t)$

$= \frac{1}{2}\left[\cos(\omega_c - \omega_m)t + \cos(\omega_c + \omega_m)t\right]$

(2)

iii.



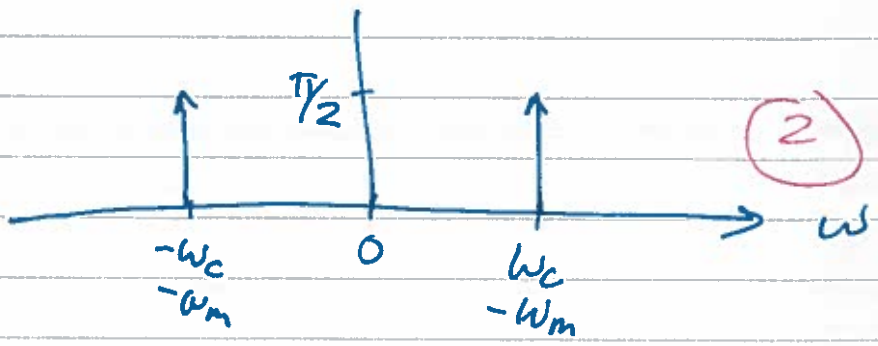
(2)

iv. The AM effectively shifts the spectrum of $\hat{m}(t)$ to the frequency range surrounding the carrier frequency $\pm \omega_c$. (1)

v.

$$\phi_{USB}(t) = \frac{1}{2}\cos(\omega_c - \omega_m)t$$ (2)

vi.



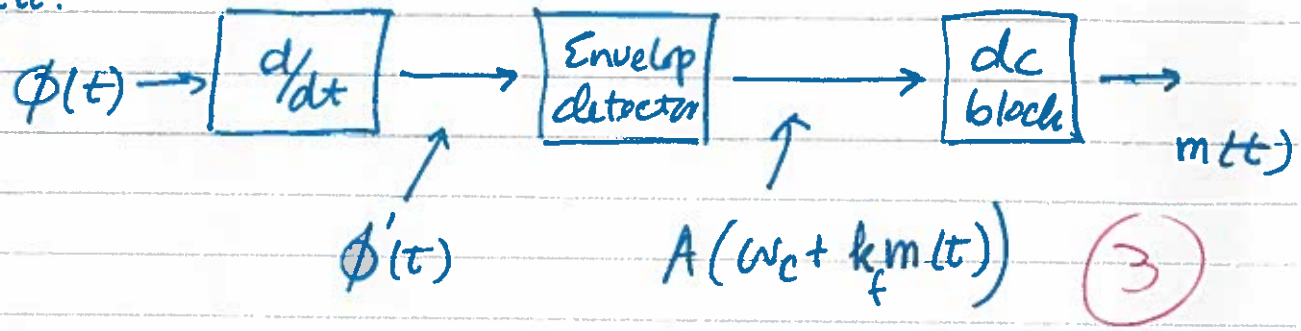
(2)

1. d. i. $\phi(t) = A\cos\left(\omega_c t + k_f \int m(\alpha)\,d\alpha\right)$ (2)

ii. $\phi'(t) = d\phi(t)/dt$

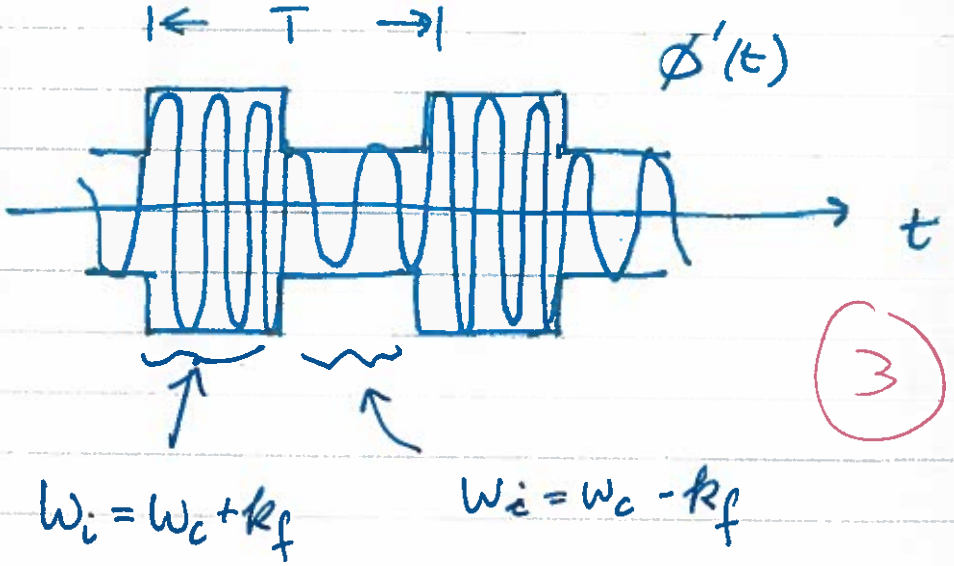
$= A\left[\omega_c + k_f m(t)\right] \cdot \sin\left[\omega_c t + k_f \int m(\alpha)\,d\alpha\right]$ (2)

iii.



(3)

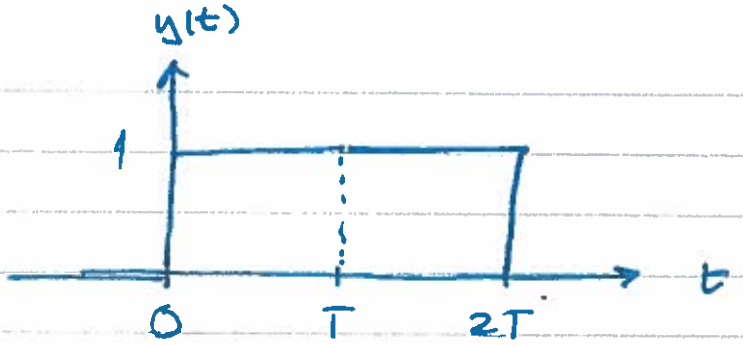
iv.



$\omega_i = \omega_c + k_f$ $\omega_i = \omega_c - k_f$

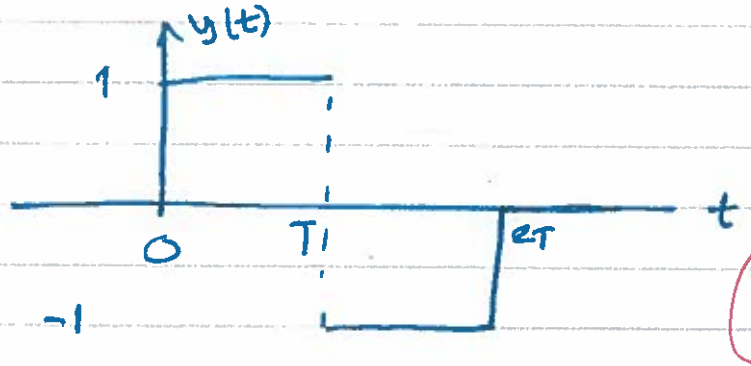
(3)

2. a i.



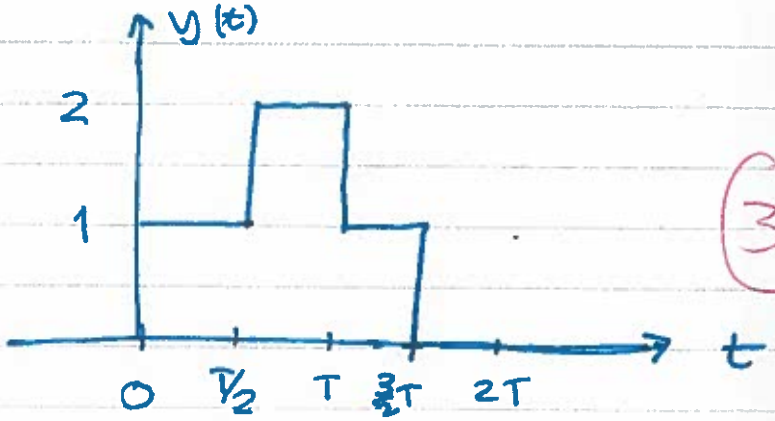
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ii.



3

iii.



3

iv. maximum signal rate :

$$f = 1/T$$

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It is so because each impulse (signal) has a response time (duration) of T time units (sec). To avoid use of complicated technique for receiving, keep the signal rate of $1/T$ samples per sec will be supported by the channel.

2. b. i. $f(t) * g(t)$

$$= \int_{u=-\infty}^{\infty} f(u)\, g(t-u)\, du$$

(3)

ii. $\mathcal{F}[f(t) * g(t)]$

$$= \int_{t=-\infty}^{\infty} \int_{u=-\infty}^{\infty} f(u)\, g(t-u)\, e^{-j\omega t}\, du\, dt$$

$$= \int_{u=-\infty}^{\infty} \int_{t=-\infty}^{\infty} f(u)\, g(t-u)\, e^{-j\omega t}\, dt\, du$$

$$= \int_{u=-\infty}^{\infty} \int_{t=-\infty}^{\infty} f(u)\, g(t-u)\, e^{-j\omega(t-u)} \cdot e^{-j\omega u}\, dt\, du$$

$$= \int_{u=-\infty}^{\infty} f(u)\, e^{-j\omega u} \int_{t=-\infty}^{\infty} g(t-u)\, e^{-j\omega(t-u)}\, dt\, du$$

$$= \int_{u=-\infty}^{\infty} f(u)\, e^{-j\omega u} \cdot \int_{t'=-\infty}^{\infty} g(t')\, e^{-j\omega t'}\, dt'\, du$$

where $t' = t - u$

$$= \int_{u=-\infty}^{\infty} f(u)\, e^{-j\omega u}\, G(\omega)\, du$$

(5)

$$= G(\omega) \cdot F(\omega)$$

2. b. iii. $F(\omega) * G(\omega)$

$$= \int_{u=-\infty}^{\infty} F(u)\, G(\omega - u)\, du$$

(2)

iv. $\mathcal{F}[f(t)\, g(t)]$

$$= \int_{t=-\infty}^{\infty} f(t)\, g(t)\, e^{-j\omega t}\, dt$$

$$= \int_{t=-\infty}^{\infty} \frac{1}{2\pi} \int_{u=-\infty}^{\infty} F(u) e^{jut}\, du\, g(t) e^{-j\omega t}\, dt$$

$$= \frac{1}{2\pi} \int_{t=-\infty}^{\infty} \int_{u=-\infty}^{\infty} F(u)\, g(t)\, e^{-j(\omega - u)t}\, du\, dt$$

$$= \frac{1}{2\pi} \int_{u=-\infty}^{\infty} F(u) \int_{t=-\infty}^{\infty} g(t)\, e^{-j(\omega - u)dt}\, dt\, du$$

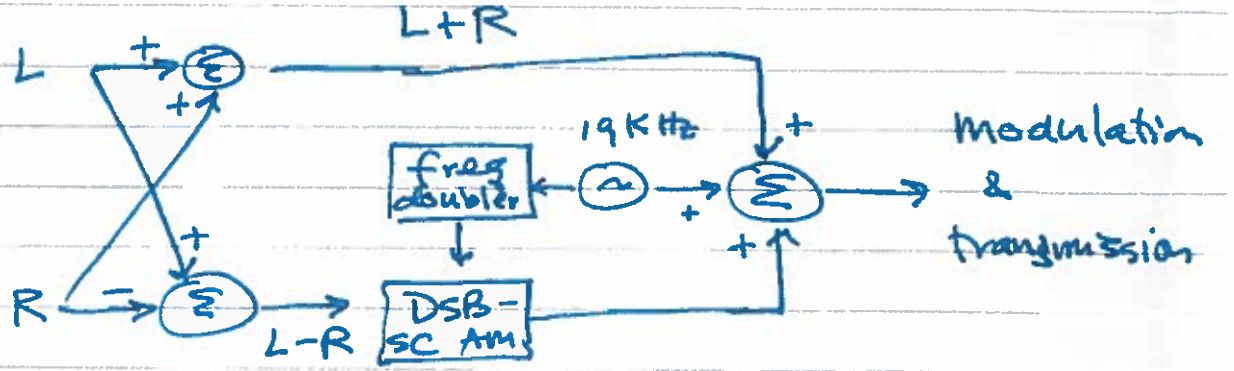
$$= \frac{1}{2\pi} \int_{u=-\infty}^{\infty} F(u) \cdot G(\omega - u)\, du$$

$$= \frac{1}{2\pi} F(\omega) * G(\omega)$$ as result in part iii. shows

(5)

(8)

3.9. i.



Form signals : L+R and L-R.

Use & double the pilot signal of 19 KHz.

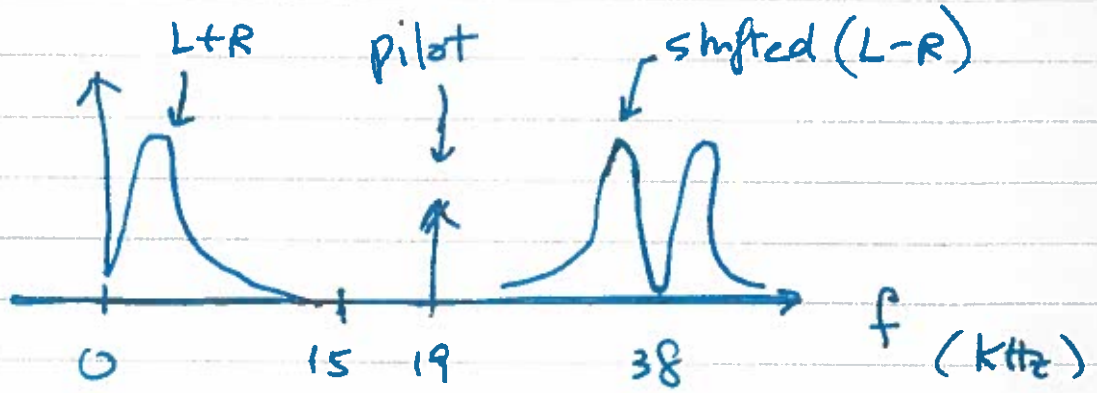
Form DSB-SC signal for L-R signal using the 38 KHz carrier.

Transmit the composite baseband signal of

L+R + 19 KHz + DSB-SC of L-R signal at 38 KHz carrier

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ii.



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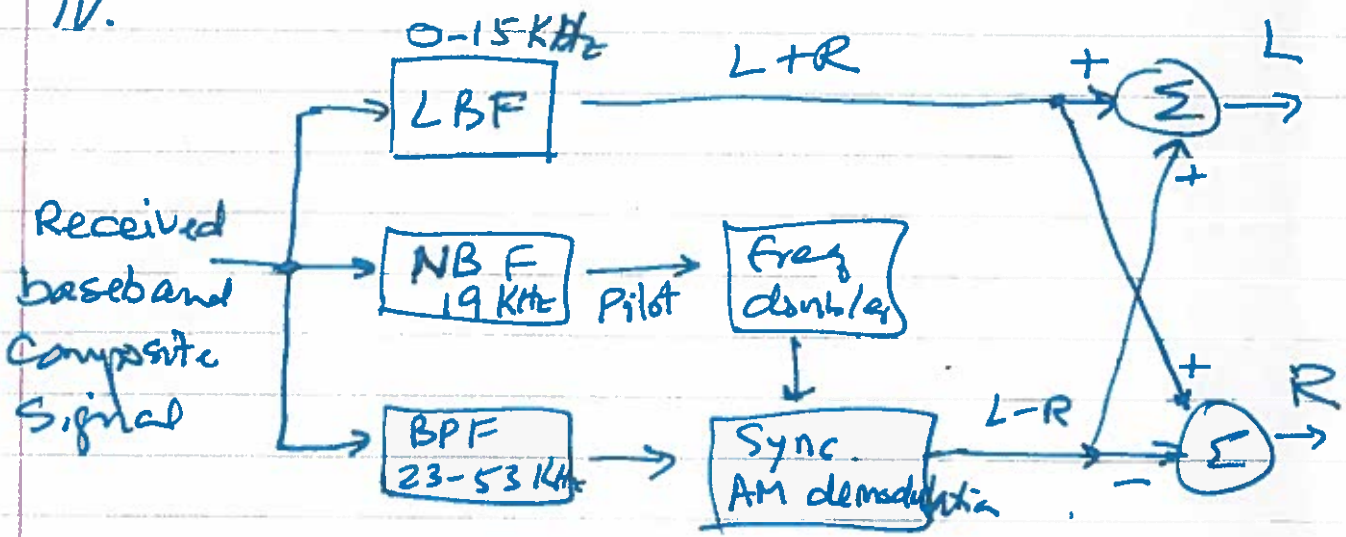
3.a. iii. The frequency of the pilot signal of 19 KHz is doubled.

Then, the 38 KHz is multiple with the L-R signal to form the DSB-SC (AM) signal at carrier frequency of 38 KHz.

The frequency multiplication is equivalent to shift the L-R signal to the carrier frequency of 38 KHz.

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iv.



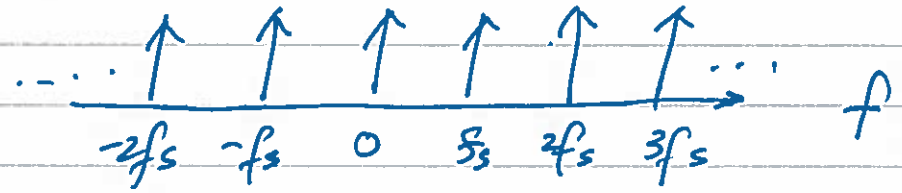
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3. b. i. Due to Nyquist sampling criterion:

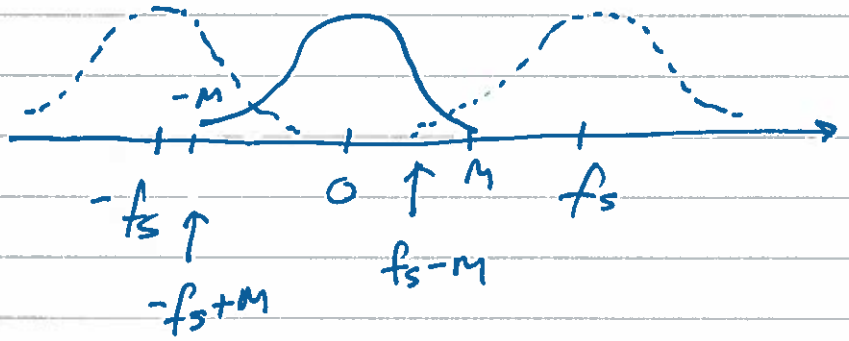
$f_s \geq 2 \times$ Bandwidth of $m(t)$

So, $M = f_s/2$

This is so because the periodic samples are equal to multiplying a train of impulses with the signal $m(t)$. The train of impulses has a spectrum of



As a result of sampling, the samples of $m(t)$ have the following frequency spectrum:



From the above diagram, in order to recover $m(t)$, we require

$f_s - M \geq M$ to avoid overlap of replica of the original signal spectrum

$\Rightarrow \quad f_s \geq 2M$

6

3. b. ii. Clearly, $C \geq M$

because the channel bandwidth must be wider than the signal bandwidth in order to avoid loss of information. (2)

iii. Since $M = f_s/2$,

we have $C \geq f_s/2$

i.e., $f_s \leq 2C$ (2)

iv. $R = 2C \log_2 M$

$= 2C \log_2 4$

$\Rightarrow R = 4C$ bits/sec (2)